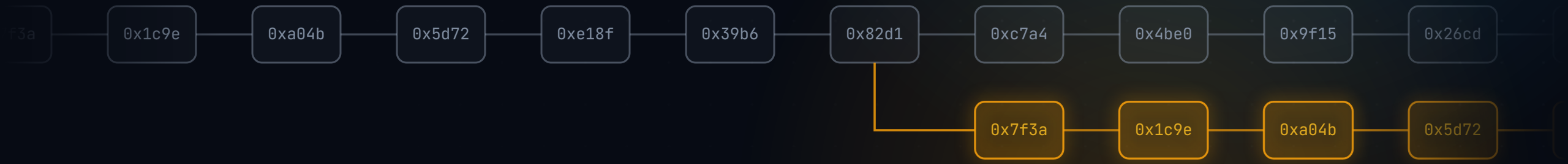
Gaming Ethereum

How Trading Bots Cut the Line and Threaten Consensus



Papers We Love Zürich

Simran Tinani · 29 October 2026

Daian et al., “Flash Boys 2.0”
IEEE S&P 2020

THE SOURCE PAPER

Flash Boys 2.0: Frontrunning in Decentralized Exchanges, Miner Extractable Value, and Consensus Instability

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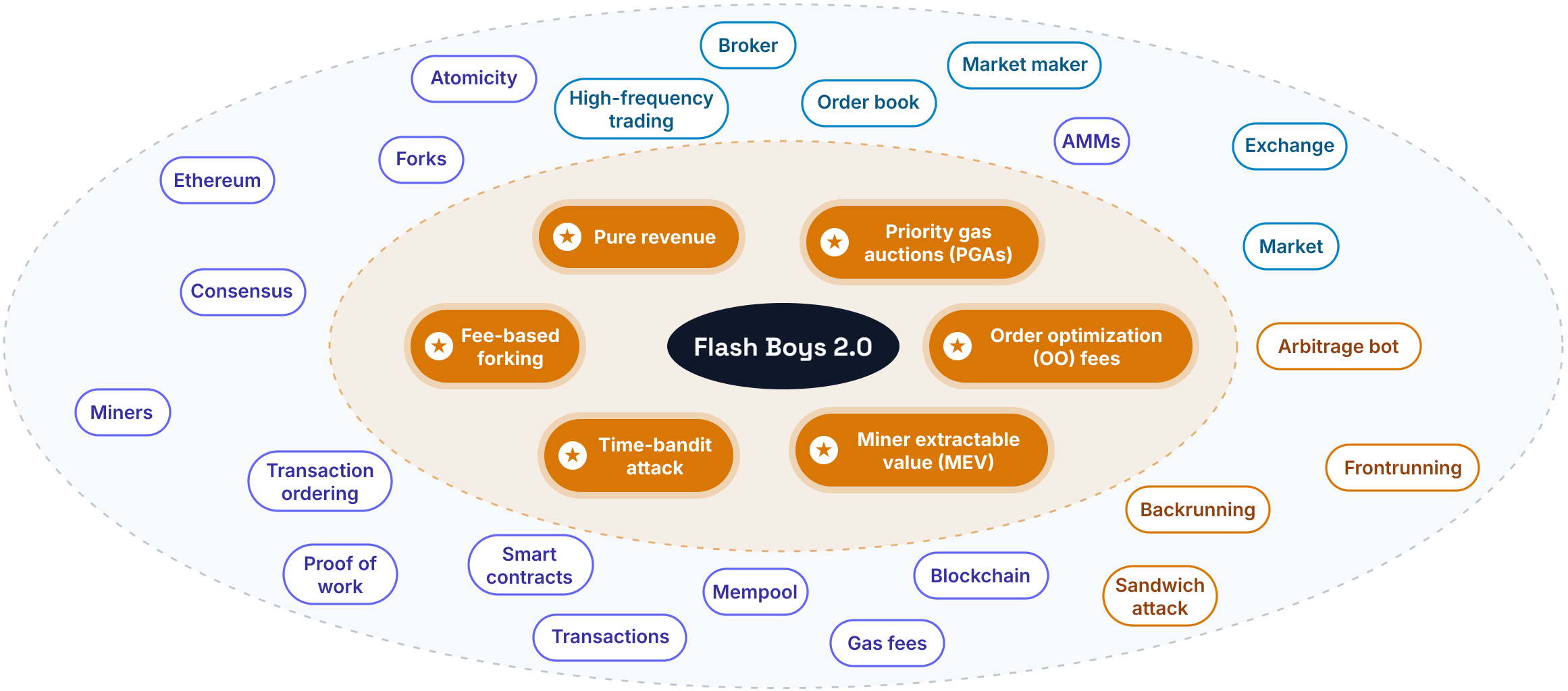
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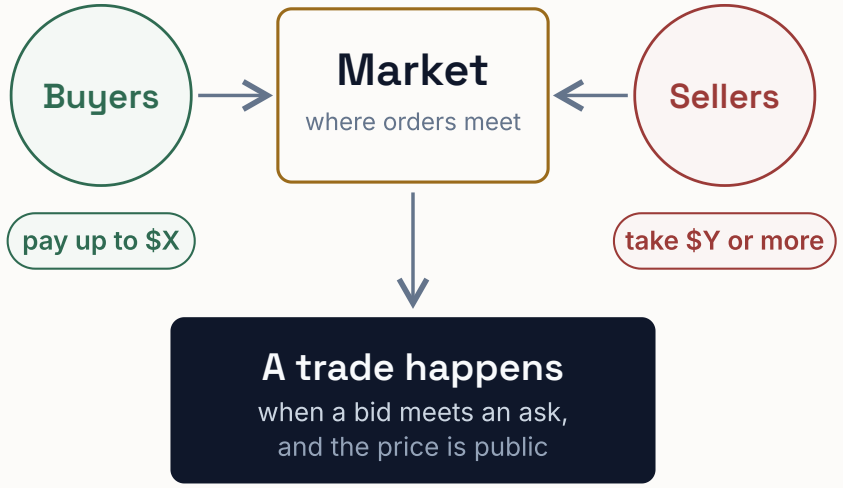
Daian et al. (IEEE S&P 2020)

Big Picture


Broker
Atomicity
Market maker
High-frequency trading
Order book
AMMs
Exchange
Forks
Ethereum
Pure revenue
Priority gas auctions (PGAs)
Market
Consensus
Fee-based forking
Flash Boys 2.0
Order optimization (OO) fees
Arbitrage bot
Miners
Time-bandit attack
Miner extractable value (MEV)
Frontrunning
Transaction ordering
Backrunning
Proof of work
Smart contracts
Blockchain
Sandwich attack
Mempool
Transactions
Gas fees

What Is a Market, and Why Do People Come?

Buyers and sellers meet to trade, and the market tells everyone what the thing is worth.



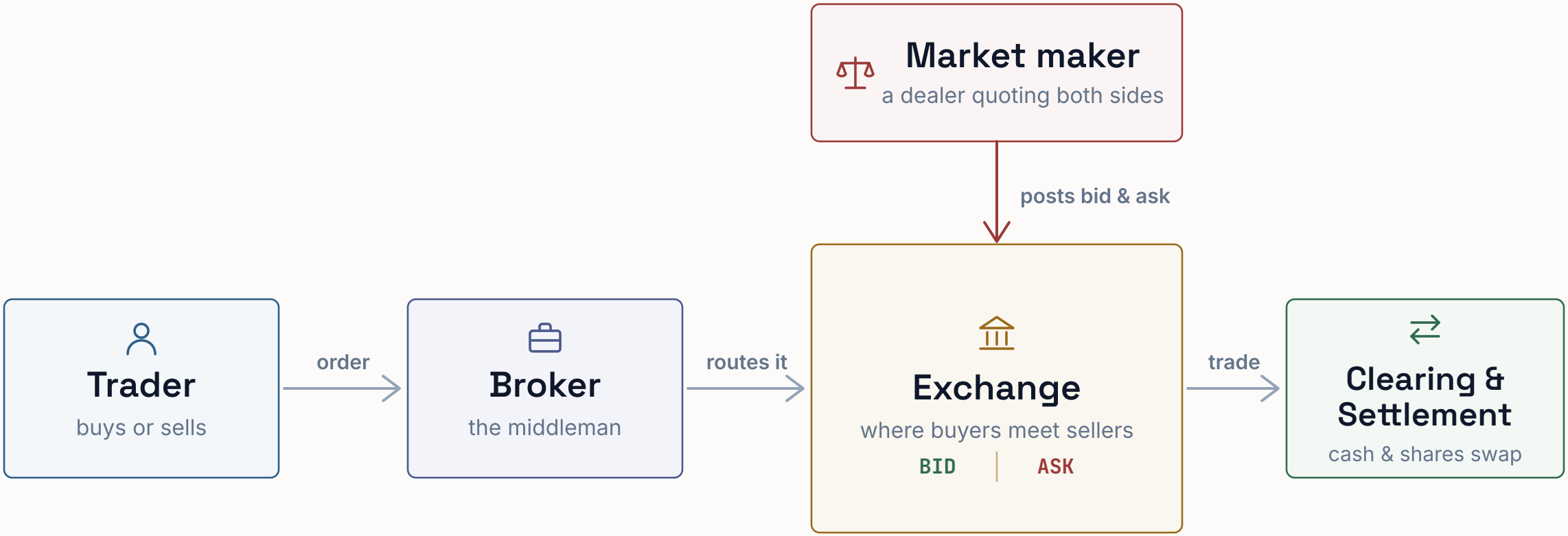
WHY PEOPLE TRADE

Invest	Turn savings into assets that can grow.	NEED
Borrow	Raise cash now and repay it later.	NEED
Hedge	Lock in a price to reduce a risk you already have.	NEED
Exchange	Swap one asset for another, such as cash for shares or one token for another.	NEED
Speculate	Bet on information or a view that others do not have.	PROFIT
Arbitrage	Profit when the same thing is priced differently in two places.	PROFIT

MARKETS BY WHAT IS TRADED

- **Stocks** — the Flash Boys setting
- **Crypto tokens** — this paper's setting
- **Currencies** — dollars, euros
- **Bonds** — tradable loans
- **Commodities** — oil, gold, wheat

Who Is Involved in a Trade?



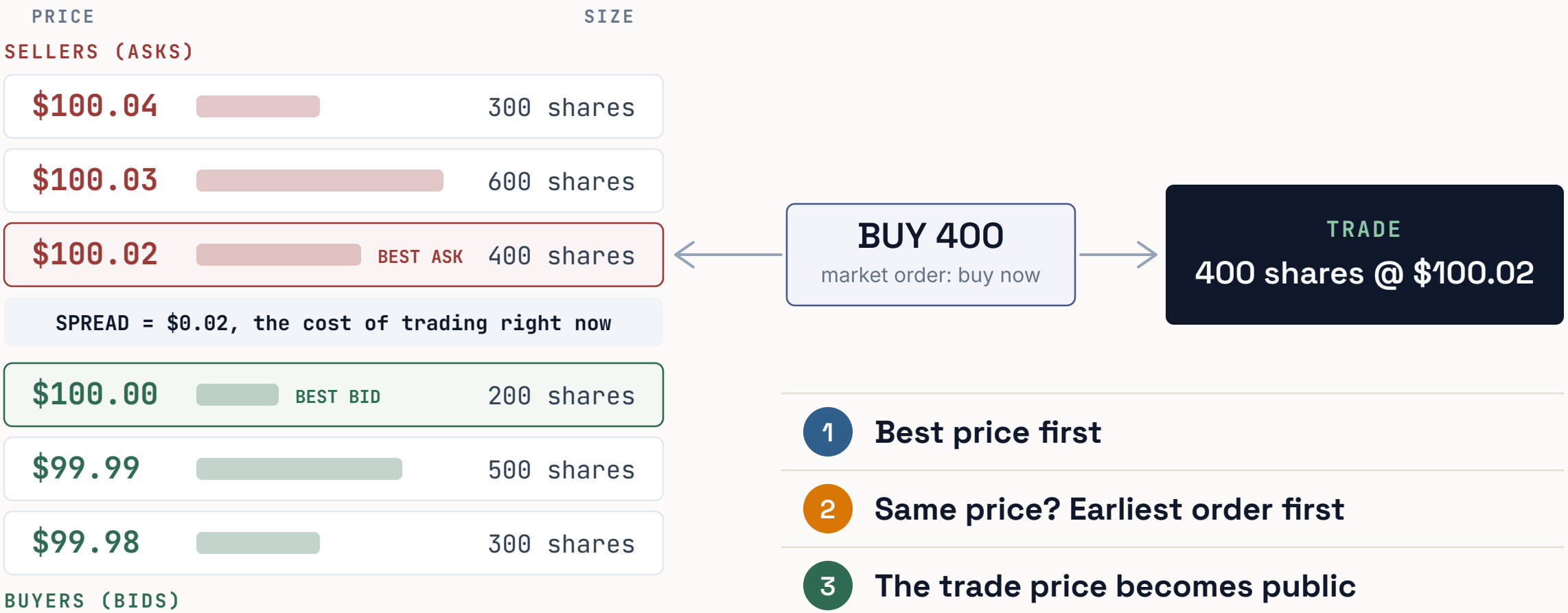
BID (BUYER)
“I’ll pay up to $X”

ASK (SELLER)
“I’ll take $Y or more”

When X meets Y, a trade happens
and that price is what everyone else sees.

TRADFI BASICS 3 OF 7

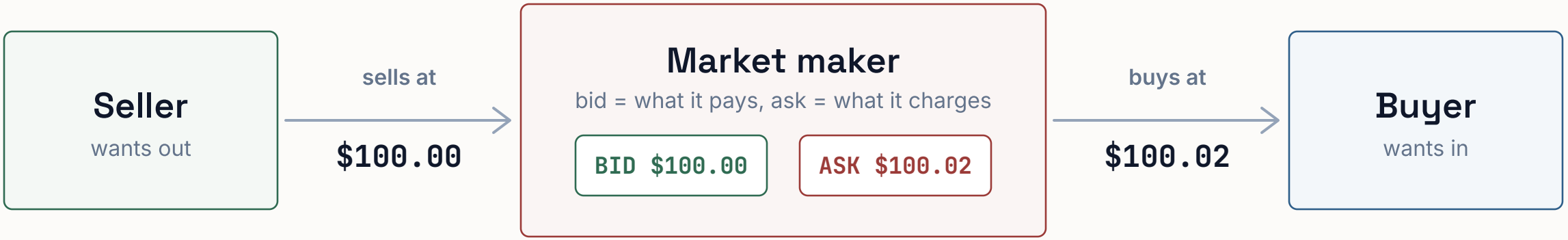
The Order Book: How Prices Are Matched



Every price on the left is a **limit order**: a standing offer anyone can post. The buyer sends a **market order**, which takes the best offers available. The book is anonymous: the other side could be a market maker, an investor or a fund.

What Is a Market Maker?

Always willing to buy and sell. That keeps a market **liquid**: easy to trade quickly, in size, at a small cost.



Finding someone who wants the exact opposite trade, right now, is slow.
The spread is the price of trading instantly.

It holds inventory, so it carries risk.
If the price moves against it, it can **lose more than the spread earns.**

In practice: mostly electronic firms like Citadel Securities and Virtu, but not every HFT is one. In DeFi, liquidity providers.

Why Are There Multiple Exchanges?

ONE STOCK, MANY VENUES	BID	ASK
NYSE	$100.00	$100.03
Nasdaq	$99.99	$100.02
Cboe BZX	$99.99	$100.03
IEX	$99.98	$100.04
Best across all (NBBO)	**$100.00**	**$100.02**

1. Regulators wanted competition
2. Venues compete on fees and speed
3. So prices end up scattered

NBBO: the national best bid and offer, the best prices found across all venues.

Arbitrage, HFT and Frontrunning

Three related ways of profiting from information or speed.

Arbitrage

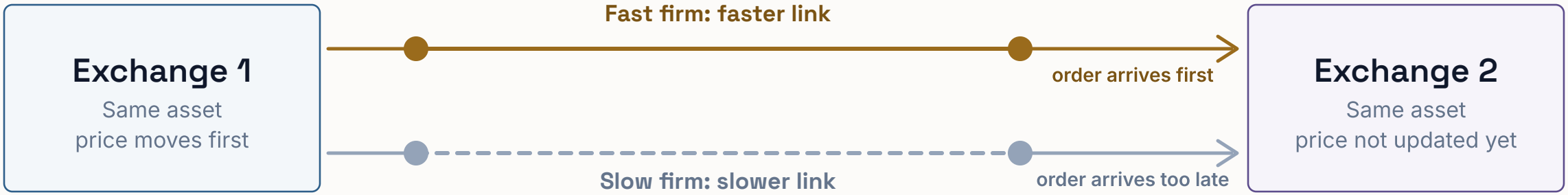
Buy where it is cheaper,
sell where it is more expensive.
Uses public prices. Legal.

High-frequency trading

Automated trading in which
speed determines profit.
Speed wins arbitrage races.

Frontrunning

Trading ahead of a known
pending order.
Illegal when a broker does it.



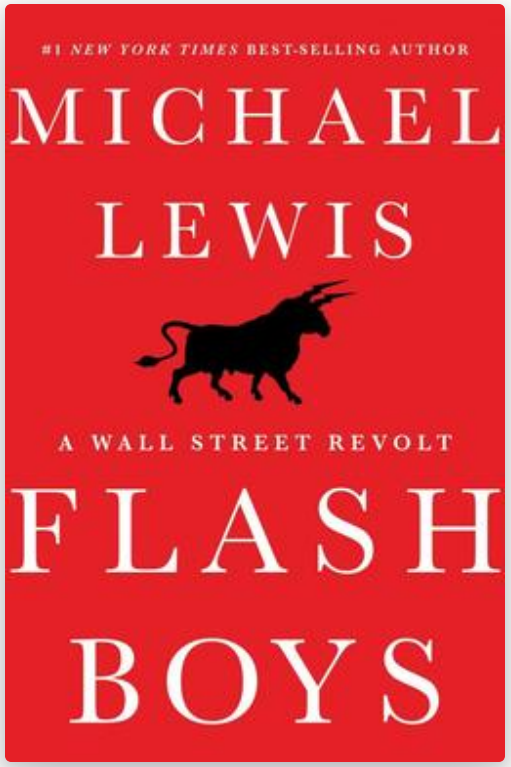
Budish, Cramton & Shim (2015): continuous limit order books give rise to this race for speed.

The Book: *Flash Boys* (2014)

Michael Lewis, *Flash Boys: A Wall Street Revolt*. W. W. Norton, March 2014.

- Regulators and researchers already knew about high-frequency trading (HFT). The book made it a mainstream public debate.
- Lewis argues that HFT firms use speed to trade ahead of investors, and calls the U.S. stock market "rigged." Brad Katsuyama's team founded the IEX exchange in response.
- Critics disputed parts of it. SEC Chair Mary Jo White told Congress U.S. markets "are not rigged."

? **Does this carry over to DeFi?**



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The World of DeFi

Financial services without a trusted intermediary.

1

Non-custodial

You control your funds at all times, with no custodian to trust or pay.

2

Permissionless

Anyone can use it, and no third party can block or censor access.

3

Openly auditable

Anyone can inspect the state of the system, for example whether it is fully collateralized.

4

Composable

Services combine like building blocks into new financial products.

Cryptocurrency

A digital currency recorded on a blockchain, such as Bitcoin or ether.

Smart contract ★

A program stored on a blockchain. It runs when a transaction calls it.

Token

A digital asset issued on a blockchain, often through a smart contract.

Stablecoin

A token designed to hold a steady value, usually $1.

Wallet

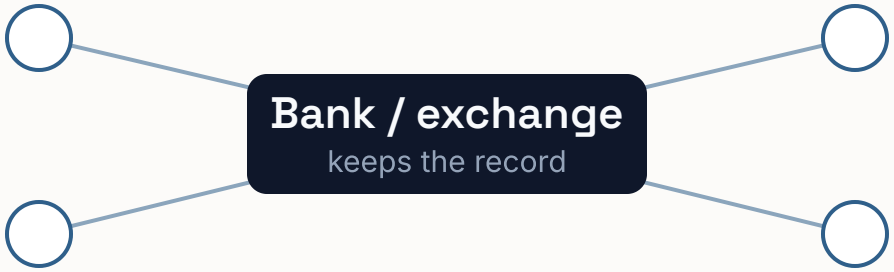
Software that holds the keys that control your funds.

What Is a Blockchain?

A shared record that many independent computers keep adding to and agree on, with no single trusted owner.

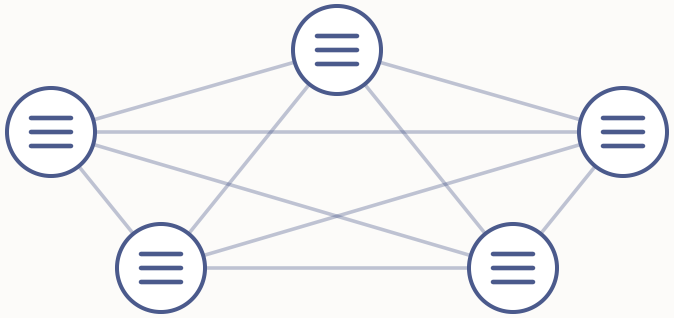
Central authority

Users must trust one party.



Validators

Each keeps a copy and checks the others.



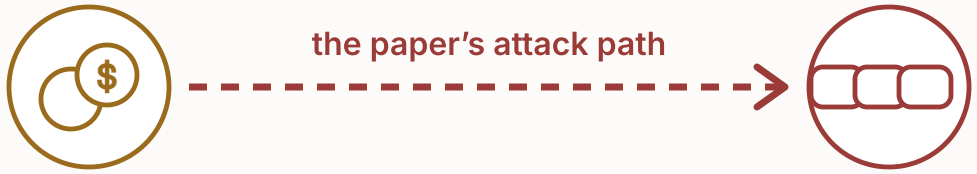
How do you replace a central intermediary?

With a network whose incentives reward following the protocol.

Blockchain

Computer science	Market microstructure	Cryptoeconomics	Political philosophy
"Is execution valid?"	*"Is clearing fair?"*	*"Are incentives aligned?"*	*"Is the network neutral?"*

Where Blockchain Security Can Fail



Code

A bug in the contract's logic lets someone make it do something its authors did not intend, often draining funds. The chain itself works as designed.

Example: the 2016 DAO hack. The contract was called again before its balance updated.

Incentives

The code is correct and the chain follows its rules, but the rules pay for harmful behaviour. Whoever pays more fees gets their transaction ordered first.

Examples: frontrunning and priority gas auctions. These hurt users, and the paper shows the fees can grow large enough to hurt the chain.

Consensus

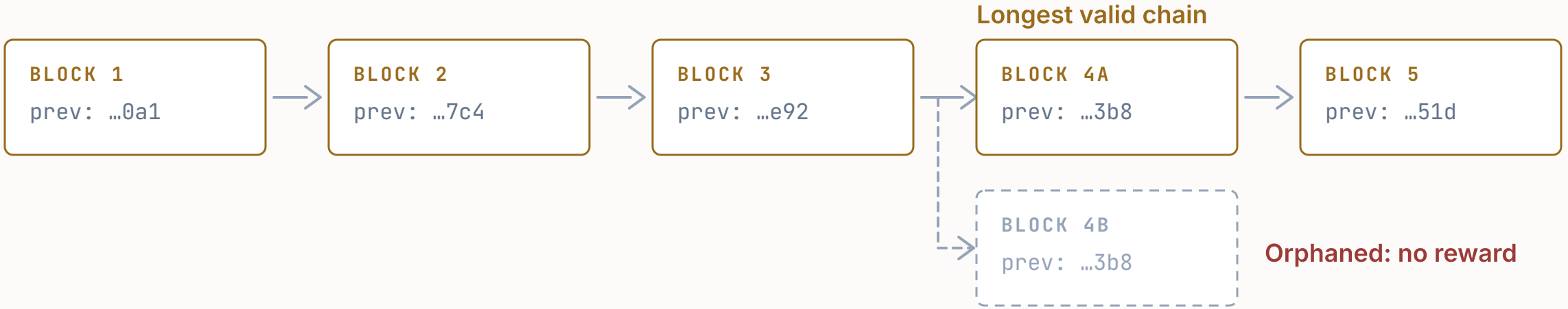
Miners or validators rewrite or reorder past blocks, so the record that everyone agrees on can change.

Examples: time-bandit attacks and undercutting. A miner is paid to fork the chain when ordering fees in old blocks exceed the block reward.

BLOCKCHAIN BASICS 3 OF 9

Bitcoin

2009: a digital currency with no bank or company in charge. Its scripts are deliberately limited, with no loops.



Nakamoto consensus

Who decides the next block?

Miners compete. Everyone follows the longest valid chain, so when two blocks appear at once, the longer chain wins.

Reorg

A longer chain replaces old blocks.

A node that sees a heavier chain switches to it, so a confirmed transaction can be undone.

Proof of Work

How strangers agree on who adds the next block.

Hash(header) ≤ target

HEADER =

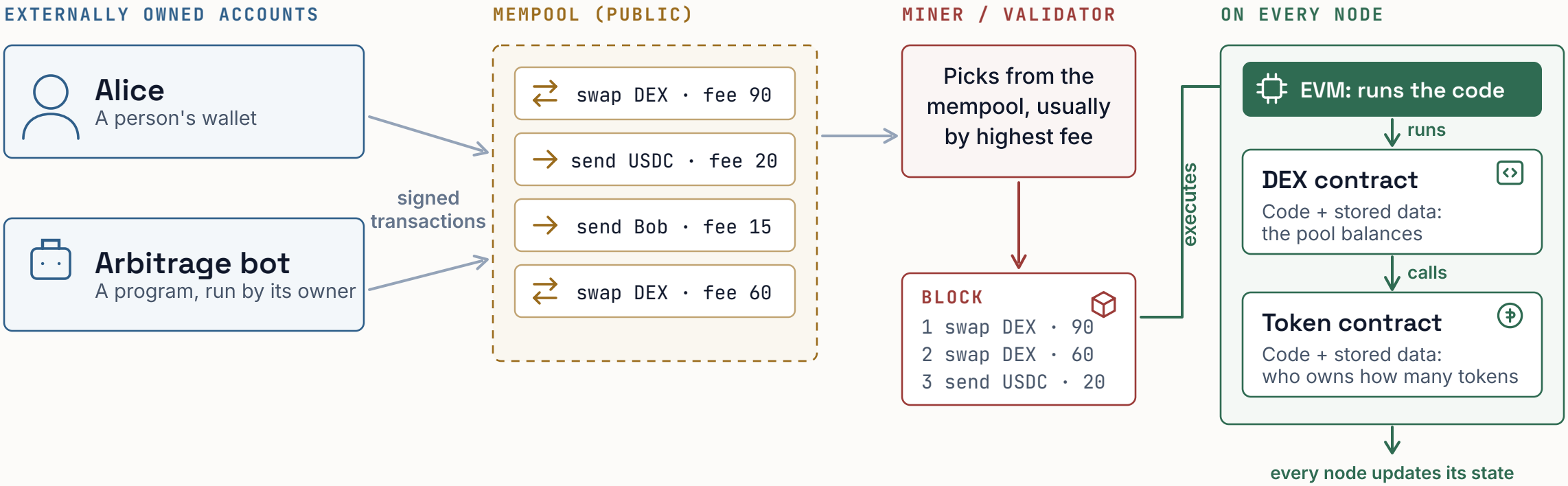
version	prev_hash	merkle_root	time	bits	nonce
protocol version	previous block's hash	fingerprint of transactions, changed by the extra nonce	timestamp	encodes the target, retuned every 2016 blocks	32 bits, tried one by one

Two numbers to vary: the nonce, and an extra nonce in the coinbase, the miner's own reward transaction.

- Miners guess nonces until the hash is low enough.
- The first to succeed gossips the block to its peers. Anyone can verify it.
- Rewriting history means redoing the work.
- Only blocks that stay in the chain get paid.

Ethereum: A Programmable Blockchain

Bitcoin tracks coin balances. Ethereum also stores and runs programs.



Nodes relay transactions peer to peer, so mempools differ, and some go straight to a miner. The paper's bots watch the public one.

Smart contracts let applications with complex logic, like exchanges, lending and escrow, run on-chain.

Decentralized Exchanges (DEXes)

An exchange run by smart contracts. No company holds your funds, unlike Coinbase or Binance.

Order book

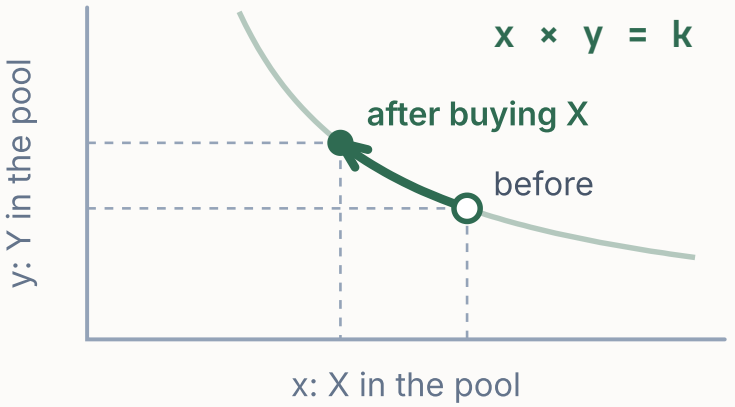
Buyers and sellers post bids and asks. A trade happens when they match.

Orders and cancellations are transactions too, so a bot can fill an order just before its cancel lands.

Examples: EtherDelta, 0x

Automated market maker (AMM)

Traders swap against a pool of two tokens, and a formula sets the price.



One X costs y / x of Y. Buying X removes X and adds Y, so the price rises.

Examples: Uniswap, Bancor

Aggregator

Splits a trade across DEXes to get the best price.

Holds no liquidity of its own.

Example: 1inch

A Transaction's Journey

transaction

nonce	to	value	data	gas limit	gas price
order number	recipient address	ETH sent	function and arguments	most gas to use	fee per gas unit

signature

Sign and send

The sender signs with a private key and broadcasts it.

Wait

Nodes check it and hold it in the public mempool, where **anyone can see it**.

Get selected

Within the block's gas limit, a miner or validator picks the highest-paying.

Where front-running happens

Run

Every node runs it in the EVM. If it fails, changes roll back but the gas is still paid.

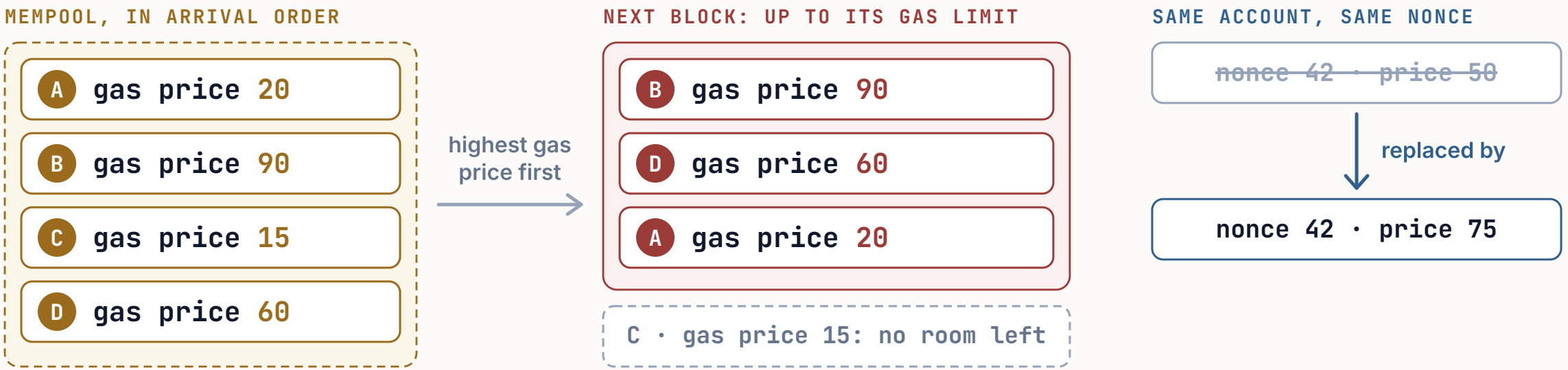
Confirm

It is in a block. Each block built on top makes it costlier to reverse.

Transactions are atomic: one runs completely or not at all, so a contract can chain several trades and revert if they do not pay. Transactions run one at a time, so their order decides the outcome.

Gas Price Bidding

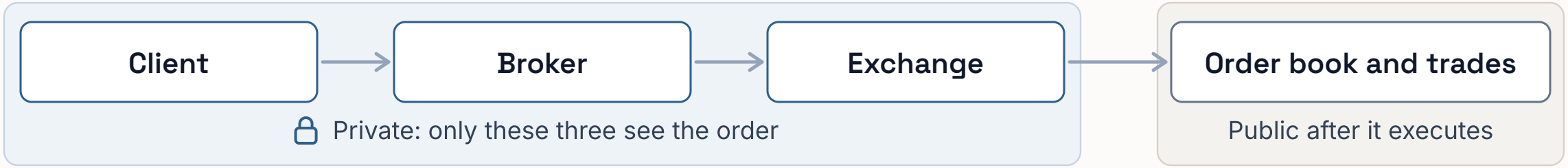
Users bid for limited block space.



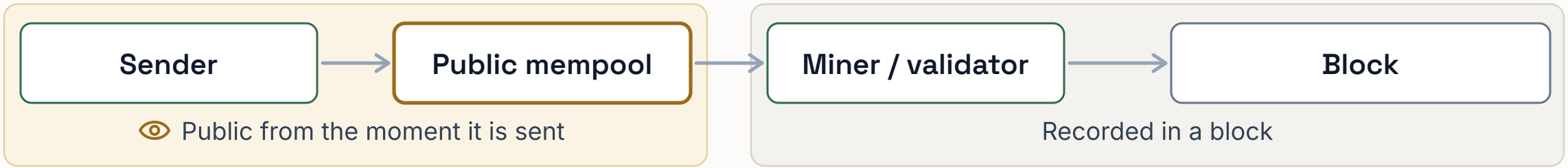
- Highest gas price goes first, until the block's gas limit is full.
- **Account nonce** (not the mining nonce) counts transactions. Same nonce, higher bid replaces.
- Too low a bid waits, and may be dropped if the mempool fills.

The Public Mempool

TRADITIONAL MARKET



ETHEREUM



- Transactions are broadcast before any block includes them, so anyone can read them.
- Fees decide the order, so a bot can copy a profitable transaction and pay more to go first.
- Bots bid against each other, and the miners collect the fees.

Arbitrage on Ethereum

TRADITIONAL: TWO SEPARATE ORDERS

Buy at 2 ETH on exchange 1

the price can move

Sell at 3 ETH on exchange 2

Risk
One trade can succeed while the other fails.

ETHEREUM: ONE TRANSACTION

THE BOT'S SMART CONTRACT

Buy at 2 ETH on DEX 1

Sell at 3 ETH on DEX 2

Risk-free
The contract's code reverts everything if a trade fails.

★ **Pure revenue:** the paper's term for trades issued atomically through a smart contract that profit unconditionally in every traded asset. The authors measured over $6M of it, a **lower bound**: only five DEXes, and bots with at least 4 bids, were counted.

Front-Running

Front-running means trading ahead of a pending order you can see.

Strategy 1: Trade ahead of a user

A bot can wrap a user's trade. Pool: 1,000 X and 1,000 Y, with x × y = k = 1,000,000 fixed.

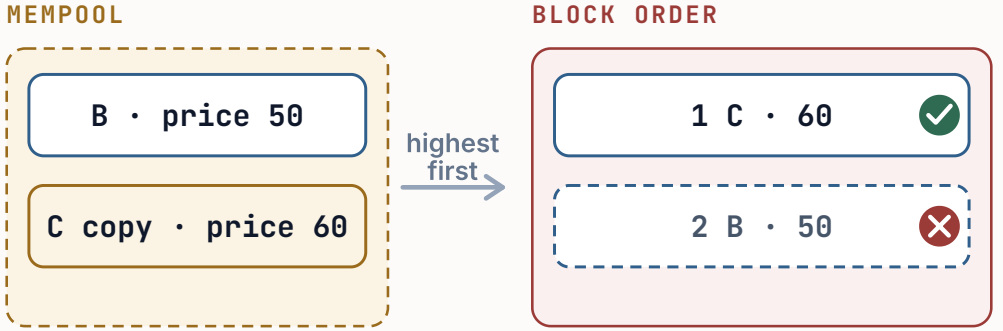
#	TRADE	GAS PRICE	Y
1	**Bot buys 50 X**	51	pays 52.6 k/950 - 1,000
2	**User buys 100 X**	50	pays 123.8 k/850 - k/950
3	**Bot sells 50 X**	49	gets 65.4 k/850 - k/900

Without the bot, the user pays 111.1 Y (k/900 − 1,000).

Result: bot gains 12.7 Y, user pays 12.7 Y more.

Strategy 2: Copy a bot

A bot copies another bot's profitable trade and bids a higher gas price.



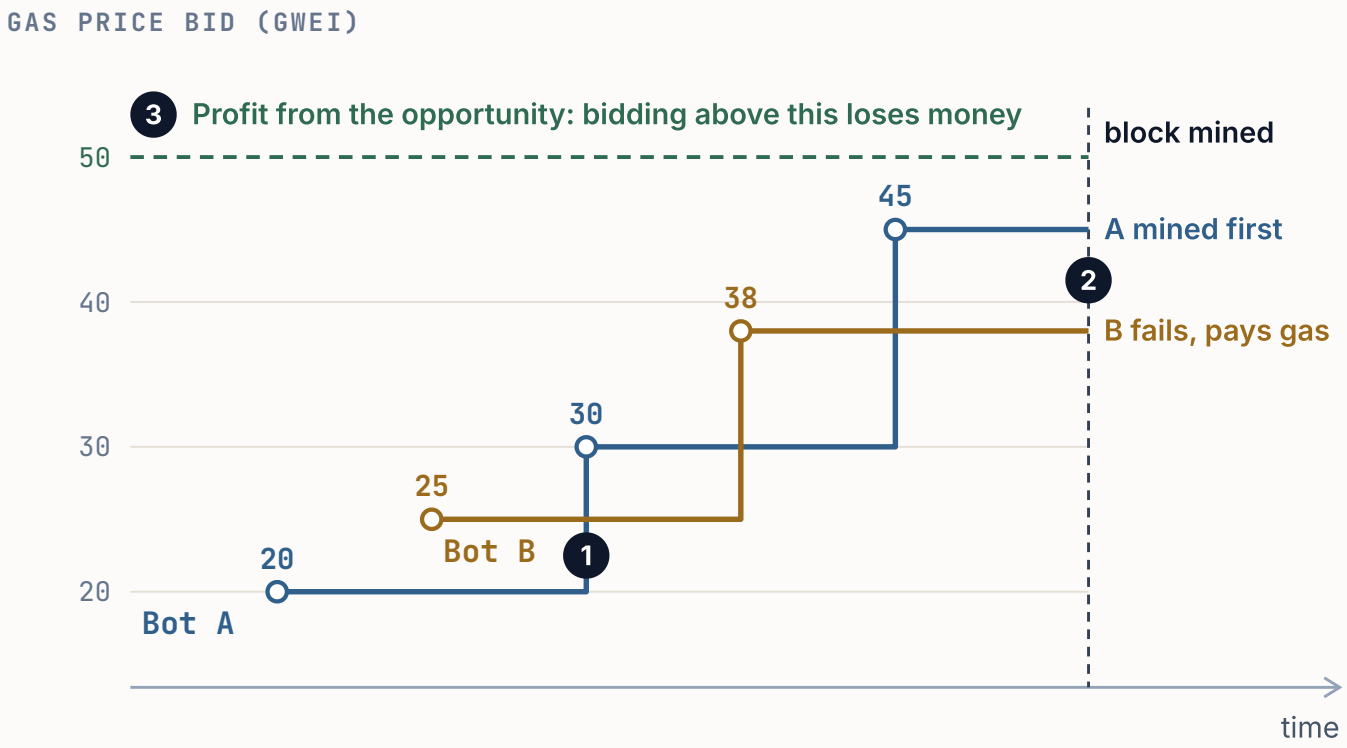
C takes the profit.
B finds it gone, still pays gas.

1. **Bot B** sends an arbitrage, gas price 50.
2. **Bot C** copies it with gas price 60.

Result: the higher gas price goes first.

Priority Gas Auctions

When several bots see the same opportunity, they bid against each other in the open mempool.



Illustrative numbers. In the paper's TokenStore auction, the two bots sent 42 and 43 bids in about 13 seconds (Daian et al., Fig. 2).

1 Rebid

Send a new transaction with the same nonce and a higher gas price. It replaces the old one.

2 Highest wins

The miner includes the highest gas price first. The other bot's trade fails, and it still pays gas.

3 Profit is the cap

Bidding above the opportunity's profit loses money, so rebidding stops there.

Why PGAs matter: priority fees are one form of miner extractable value, which the paper calls a systemic risk to consensus-layer security. Losing bids also land on chain as failed transactions that use block space (later surveys).

Game Theory in Brief

A tool for modelling how rational players, each maximizing its own payoff, act when their incentives interact.



Both would prefer 3, 3, but neither can gain by moving alone from 1, 1.

Illustrative payoffs: a prisoner's dilemma.

Nash equilibrium

No player can do better by changing only their own strategy, meaning their plan for how to play.

Dollar auction

The winner gets the prize, but losers pay their last bid too, so bids can escalate. A losing bot pays gas for its failed transaction.

Advantage

The paper compares two strategies by how much better one does against the other.

Role for blockchains: nodes can check that blocks are valid, but not which valid blocks miners build on or how they order transactions. Incentives govern those choices, and game theory asks when deviating pays.

Modeling a Priority Gas Auction

Modeled as a “dollar auction”: bidders compete for $1 in continuous time, and a block ends it at a random time.

Δ Imperfect information

A player sees rivals’ bids only after a fixed delay, its latency Δ.

ℓ All-pay

The loser still pays gas for its failed transaction: ℓ(last bid), usually less than the bid.

δ Rate limit

A player waits δ between raises.

ι Minimum raise

A replacement must raise the price by a fraction ι. The Parity Ethereum client requires 12.5%.

λ Random block time

Mining is memoryless, so the wait for a block is exponential with rate λ.

$$P(\text{block by } t) = 1 - e^{-\lambda t}$$

Strategy

From the bids seen so far, the time and its state, it returns an action (bid or none), a wake time and a new state:

$$(a,\, t_w,\, \sigma') \leftarrow S_i(b^*,\, t^*,\, \sigma)$$

Payoff

$\mathrm{PO}(S_0, S_1)$ is the expected profit of S_0 against S_1; an average, since the block time is random.

Advantage

$$\mathrm{Adv}(S_0, S_1) = \mathrm{PO}(S_0, S_1) - \mathrm{PO}(S_1, S_0)$$

If positive, S_0 has the advantage.

Null-profitable (non-dumb)

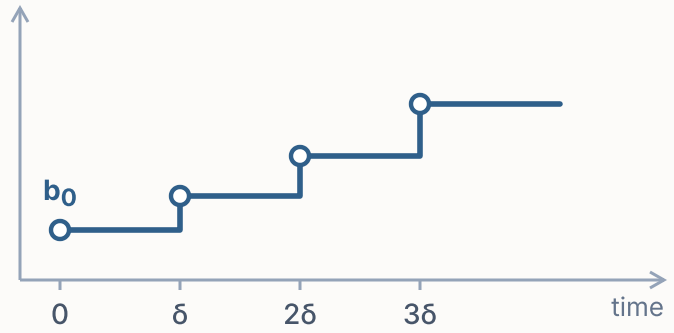
$\mathrm{PO}(S_0, S_\varnothing) > 0$, where $S_\varnothing$ never bids. Equivalently, its expected winning bid is below the $1 prize.

Blind Raising vs. Counterbidding

A blind schedule can beat reacting to rivals when the reaction delay is long.

Blind raising

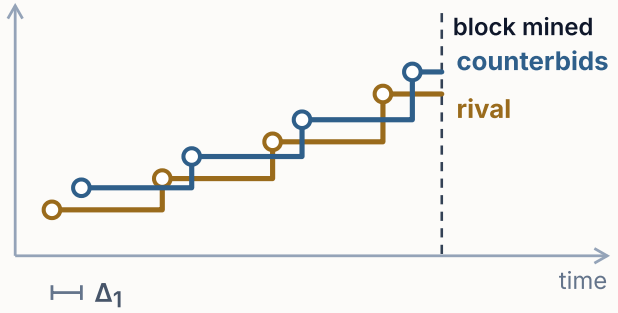
Ignores rivals. Raises its bid by a fixed fraction f every δ.



Bids $b_0\,(1+f)^k$ at time $k\delta$

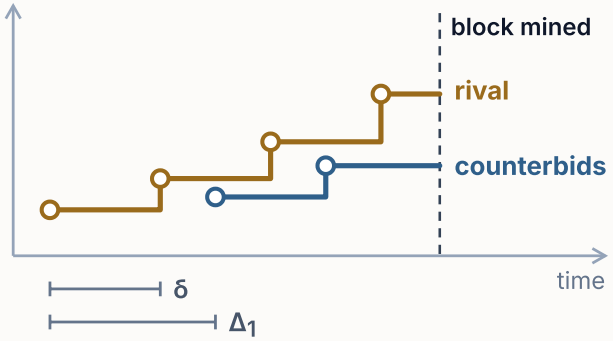
Counterbid, no delay

Sees rival bids at once ($\Delta_1 = 0$).



Counterbid, long delay

Sees rival bids late ($\Delta_1 > \delta$).



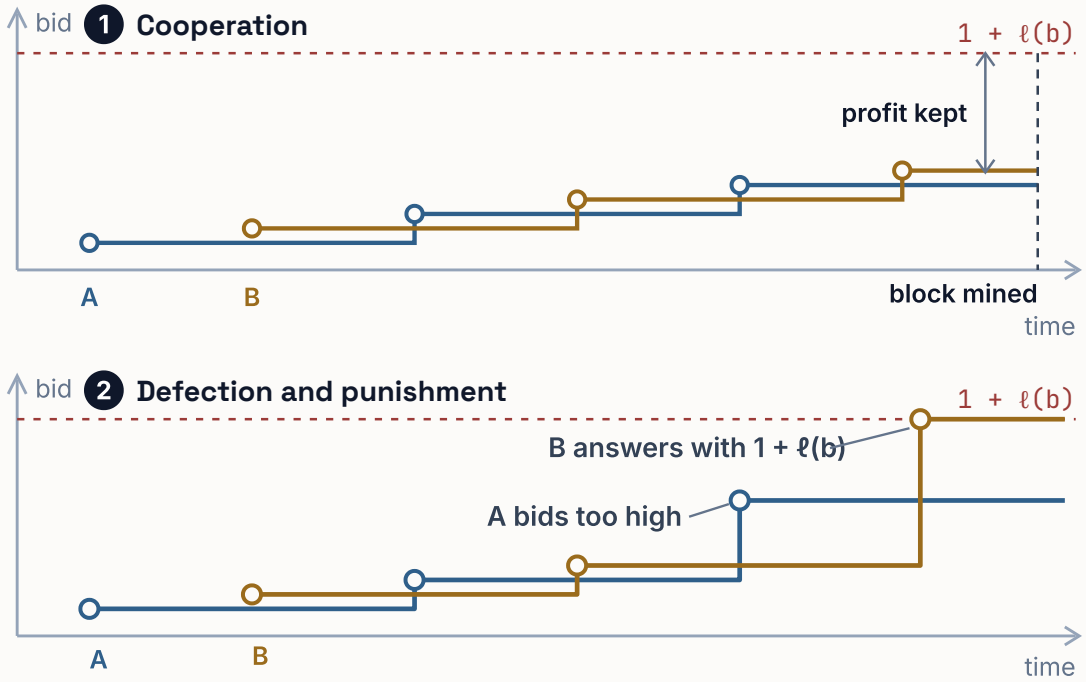
Counterbidder: bids exactly the minimum raise above the rival's latest bid it has seen (Δ_1 is its latency), but never above $1 + \ell(b_1)$, where b_1 is its own last bid. Past that, it stops and pays $\ell(b_1)$.

Observation 1. With no delay ($\Delta_1 = 0$) and a small enough loss ℓ, counterbidding beats a null-profitable blind raiser.

Observation 2. With a long delay ($\Delta_1 > \delta$), some null-profitable blind raiser has $\mathrm{Adv} > 0$ against any pure counterbidder.

Cooperation and Grim Trigger

Bidding against each other burns the profit, so can bots do better by taking turns?



1. **Cooperate.** Players take turns at agreed times, each raising by the minimum, so the profit is kept.
2. **Defect and punish (grim trigger).** A player bids early or too high. The other bot immediately bids 1 + ℓ(b): the reward (1) plus the loss ℓ(b) of its own last bid b. That is the most it would ever pay, and it leaves no profit. “Grim trigger” means punishing with no forgiveness.

Who wins. A bot wins if the block lands in its turn. Block times are exponential, so early turns are slightly likelier.

Observation 3

With this strategy, the best bids rise by exactly the minimum: each bid is the last one × (1 + ι).

Theorem

For parameters consistent with Ethereum PGAs, a **grim-trigger Nash equilibrium** exists for 2 players. The authors do not claim bots actually play it.

In other words: taking turns is a best reply to taking turns. Unlike a one-shot prisoner’s dilemma, a deviation is punished.

Measured: turns ≈ 0.4 s, latency ≈ 0.1 s, block time ≈ 15 s. Longer delays break it.

Miners Want a Slice of the Pie

Miners control transaction ordering, so they can claim the profit bots compete for.

One arbitrage transaction: buy 100 tokens for $100 on exchange A, sell them for $110 on exchange B. **$10 profit** before gas.

Through a PGA

Bots bid up the gas price; the miner collects the fees.

TRANSACTION	GAS PAID	ARBITRAGE PROFIT
Bot A succeeds	$8	$10
Bot B fails	$1	none
Miner receives		**$9** ($8 + $1 in gas)

Miner goes directly

The miner puts its own copy of the trade first.

TRANSACTION	GAS PAID	ARBITRAGE PROFIT
Miner's own copy	none	$10
Bots A and B fails	$1 each	none
Miner receives		**$12** ($10 profit + $2 in gas)

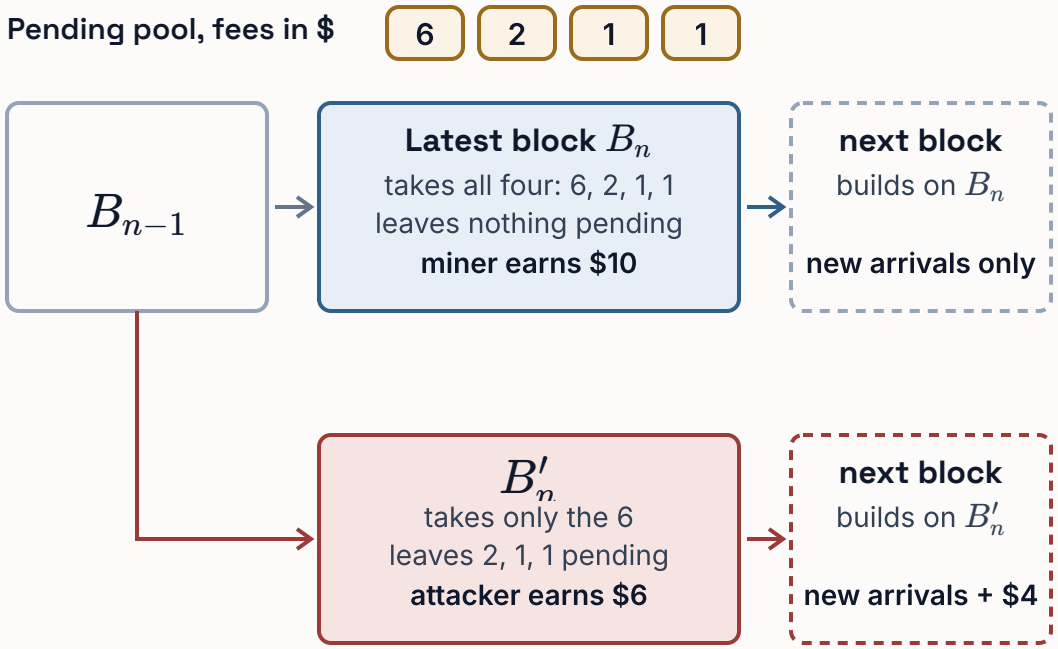
Ordering optimization (OO) fees. Value a miner can earn by controlling transaction order. PGA bids are one way it is paid.

Miner extractable value (MEV). The total Ether a miner can extract by manipulating transactions, which includes reordering them and inserting its own, over a given timeframe, possibly several blocks.

> "PGAs themselves are an indirect mechanism for miners to claim OO fees: if miners instead claimed OO fees directly, **there would be no incentive to bid in or create a PGA**."

Undercutting Attacks

A miner can fork a high-fee block, holding back some fees to attract other miners to build on the fork.



New transactions keep arriving on both chains, but only B'_n also has $4 left over, so B_n is orphaned.

Fork. A miner builds on an earlier block instead of the latest one, creating a competing chain. Only one chain survives. On the other, blocks are orphaned: their reward and fees are lost, and their transactions return to the pending pool.

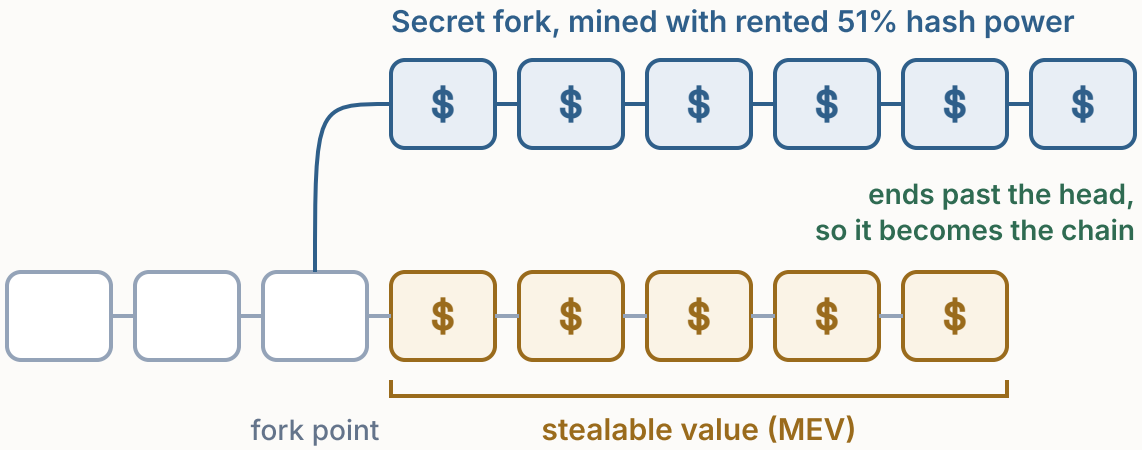
Undercutting. A miner forks the latest block and re-mines it, but takes only part of the pending fees. The rest stay unclaimed, so the fork offers the next miner more than the original chain does, and miners build on the fork.

> **Undercutting can use any MEV, not just explicit fees. In the paper's highest-revenue block, one arbitrage earned 101.6 ETH, while explicit fees were 0.022 ETH.**

No attack was observed: the threat is argued from the fees.

The Time-Bandit Attack

A miner can rewind and mine a longer fork if the value in those blocks outweighs the cost of the attack.



EXAMPLE (PAPER)

A token on an on-chain exchange spikes from $1 to $3. A miner rewrites the last 24 hours so it is on the buy side of every trade, buying at $1 before the spike.

Undercutting forks only the latest block and needs other miners to follow the unclaimed fees.
A **time-bandit attack** rewinds many blocks, needs no one to follow, and takes the MEV already in them directly.

> Ethereum DEX trading volume peaked at **≈ $1.5B** in July 2018, **more than 25 times** the **≈ $56M** estimated cost of a month-long 51% attack (March 2019). How much of it is stealable is hard to gauge.

After the Paper

Ethereum did not remove MEV. It moved the auction off-chain and made rewriting history costlier.

2019

The paper

Bots recreate Wall Street frontrunning on DEXes. They earned over $6M of arbitrage (a lower bound) and competed for it in **PGAs**. Fees that large can make forking pay: **undercutting** and the **time-bandit attack**.

2020

Flashbots Auction

Bots (called **searchers**) send **bundles**, each with a bid, over a private, off-chain channel to a Flashbots **relay**, which passes the best one to miners. Only the winner reaches the chain → no failed bids, no public gas war.

2020 on

Private mempools

Users send transactions to a private endpoint, such as *Flashbots Protect*, which passes them to block **builders**, not the **public mempool**. Bots cannot see them.

2022

The Merge

Proof of stake adds **finality**. A finalized block can only be reverted by destroying a large amount of stake, so rewinding deep history is far costlier. Recent blocks can still be reorganized.

2022

MEV-Boost

Proposer-builder separation as software, launched with the Merge. How it works is on the next slide.

Next

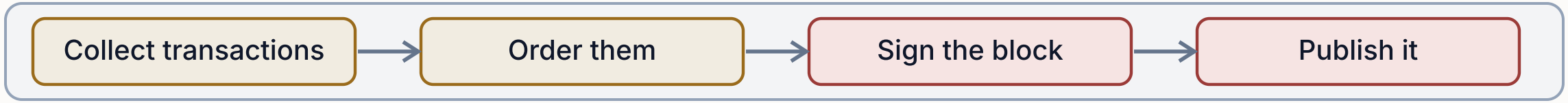
Enshrined PBS

Aims to build PBS into the protocol itself (**ePBS**, planned), reducing reliance on relays.

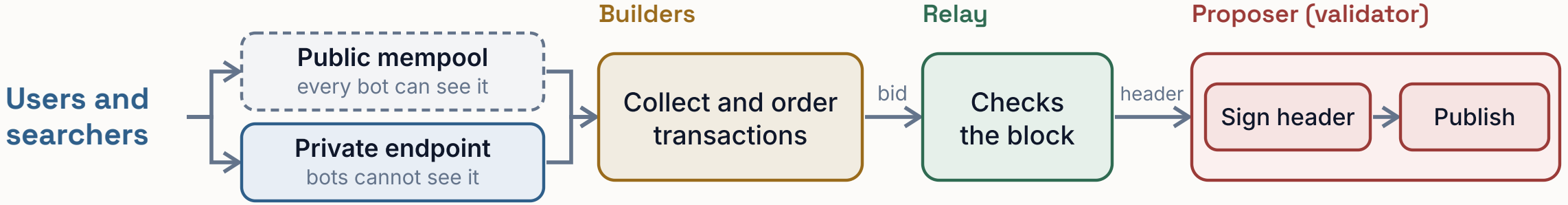
> **Frontrunning was organized, not abolished.** The public gas war became a sealed, open auction: no failed transactions, no congestion, and the profit goes to the proposer.

Proposer-Builder Separation

Before one validator does everything, on its own node, from the public mempool



After



Searchers' bots watch the public mempool. A private endpoint, such as Flashbots Protect, sends transactions straight to builders, hidden from bots.

Anyone can build. Builders compete on how much they pay the proposer, using the transactions and bundles sent to them. In practice, private order flow and capital to overbid limit who competes.

Run by firms such as Flashbots or bloXroute, off-chain, not paid by the protocol. It checks the block and holds it until the header is signed.

Any validator can opt in. It signs the highest bid's header without seeing the transactions, and builds its own block if MEV-Boost fails.

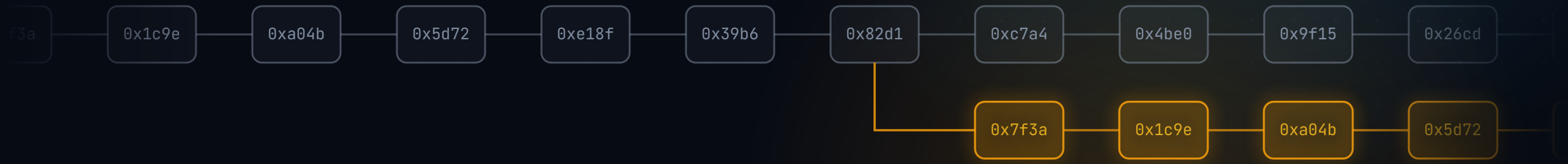
What Changed, What Did Not

MEV was institutionalized, not eliminated: it moved into an organized market.

THE PAPER'S THREAT	WHAT CHANGED	WHAT REMAINS
PGA gas wars	Bids move off-chain: bots send bundles privately, and only the winner is included.	The competition still exists, but privately, so it is much harder to observe.
Frontrunning users	Private mempools hide transactions from bots.	Builders still see every transaction, so users must trust them.
Time-bandit attacks	Finality: reverting finalized blocks needs a large slashable stake.	Recent, not yet finalized blocks can still be reorganized.
MEV itself	Captured through an open builder market, and paid to validators.	An open fix, but builders and relays now sit between users and blocks, raising censorship and trust concerns. In April 2023 a relay bug let a validator take about $25M.

Thank you

for your attention. Questions?



Papers We Love Zürich

Simran Tinani · 29 October 2026

Daian et al., “Flash Boys 2.0”
IEEE S&P 2020